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Automatic Fall Detection system for patients with Parkinson's disease

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***Abstract*— Parkinson's disease is a progressive disease that has been increasing in each year, affecting the elderly, limiting them to perform their daily activities because this progressive disease alters their gait and causes them to fall causing them strong falls causing damage, worse than they are elderly people where a fall injury can take a long time to recover. Nowadays, there are fall prevention systems that are annoying and expensive, there are even systems that are very fragile to be used in natural environments and are very complex to manipulate, therefore, it is necessary to develop a practical, simple and cheap system to prevent people with Parkinson's disease from suffering a serious accident because of the abnormal gait they may suffer. According to the proposed problem, it is intended to realize a system that prevents the fall of people with Parkinson's disease, helping patients by performing an analysis of gait variation with the motion sensor to detect and prevent falls, allowing patients to perform their activities safely by being alerted by an auditory signal at the time they have an abnormal gait. By means of the system tests, an optimal performance was visualized, obtaining 97.99% efficiency in detecting and alerting the patient.**

Keywords: BUZZER, LEDS, OMS, Parkinson, Python, Raspberry Pi, Motion Sensor.

I. Introduction

Parkinson's disease is directly related to the nervous system and worsens over time [1], the most prominent signs of this disease scrapie [2]. According to the World Health Organization (WHO), in 2018 this disease affected approximately 4.4 million people over the age of 65, and by 2035 it will affect approximately 11 million people worldwide [3], a considerable number of patients affected by this disease.

The main cause of this disease is that the neurons responsible for producing dopamine are damaged, which causes changes in the circuit of basal nuclei of the brain, in addition to the alteration of gait [4], producing symptoms that usually worsen over time or aging, increasing the chances of fall of any patient [5]. Damage from falls can affect the patient's mobility, they can even aggravate existing conditions or symptoms [6], however, patients with Parkinson's disease can respond through auditory signals when alerted [7], so that they are aware of the steps they should take and can prevent falls.

Currently the existing systems for the prevention of falls of patients with Parkinson's turn out to be invasive, expensive or uncomfortable [8], there are also a variety of fragile systems to be used under normal conditions [9], Also, accelerometers have been used as an option to evaluate patients' walking, but by influence of gravity with respect to the location of the sensor and the axis of rotation [10], it makes them unstable and unsafe for patients. Although many variables can be used to analyze the accident risk of patient walking, but the cost of this equipment is high, in addition to complex manipulation [11], so it is necessary to develop a practical, simple, and inexpensive system to prevent Parkinson's patients from suffering a serious accident when falling.

The objective of this system is to perform a fall detection analysis for patients, in such a way as to proactively support Parkinson's patients to perform an analysis of the person's gait and accident prevention, allowing patients to perform their daily activities safely. For the realization of the system, a gyroscope coupled to the body part was used to detect falls, based in the orientation so that it can be rotated without any problems together with Raspberry Pi 3 programmed in Python. Python is a language where applications are made in a simple way, being the most used by many programmers.

In section II, the exploration of the research developed. In section III, the block diagram to explain about the system. In section IV, the functionality. In section V, the results of the implementation carried out. In section VI, the discussion highlighting the essentiality of the research. Finally, in section VII, the conclusion and recommendation of the developed system is made.

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II. LITERATURE REVIEW

Parkinson's disease, in adult patients, is a complicated problem when performing their daily activities, due to the fear of suffering an involuntary fall because of this disease. For this reason, it should be forced to control in a demanding way that allows Parkinson's patients to prevent falls. For example: In [12], the authors mention that people with this disease frequently suffer from paralyzing of the gait, being a common problem caused by a motor disability that reduces the quality of life of people, in the same way it puts their lives at risk due to the falls they can suffer, hitting any part of their body, that is why they proposed to make a portable system of detection of freezing of the gait adaptable to the patient for signals of biofeedback. The methodology using a neural network so that it can learn instantly, it also uses an FPGA plate for the classification of characteristics, allowing it to adapt to the patient. As a result, a detection efficiency of 90.2% was obtained, as a conclusion this system presents a good classification performance and adapts to the patient.

In [13], researchers say that people with Parkinson's suffer from a common movement disorder that is gait freezing, presenting when the person already has Parkinson's disease in an advanced stage, causing them to have strong falls, being dangerous if it is a person of legal age where their recovery from a fracture would be very complicated, that is why decided to develop a system to detect and alert sick patients. The methodology is based on using a portable inertial sensor that will allow you to recognize the minimum latency, which together with wireless headphones will emit an acoustic signal to thaw the person's gait, preventing them from suffering a fall. As a result efficiency 91.11%, as a conclusion that detects the freezing the gait without any inconvenience on the patient with Parkinson's.

In [14], the authors mention that the Walking paralysis and tremor are ordinary signs in people with Parkinson's, making the person uncomfortable of the gait, causing patients to suffer from falls and obtain injuries that they can affect some part of their body, leading them to a limitation in their daily activities, so they decided to manage a system to detect walking paralysis in people with Parkinson's disease. The methodology is based detecting the freezing of the gait so that the functional electrical stimulation is activated by analyzing the frequency and variation in the parameters of the gait through the sensor incorporated in the system. As a result, efficiency 93.80% and as a conclusion this system uses a wireless inertial sensor that has no problems in detecting the variation of parameters.

In [15], researchers claim that Parkinson's disease fatally generates alterations in people's walking, which, if treated in the long term, can be fatal due to the many falls that the sick person may suffer, causing them to have a risk of falls involuntarily, with this, the quality of life of people is reduced due to the blow they may suffer, that is why they proposed to make a system supported by cellular phones to alert the paralyzed walking of sick patients. The methodology supported on eliminating gait freezing by means of rhythmic auditory signals by developing a time-frequency algorithm for the detection and monitoring of walking freezing in people who became ill. As a result, efficiency 91.79% and as a conclusion that keeps track of time, place, and duration of the freezing of the gait.

III. METHODOLOGY

In this section of the research, a block diagram is made to indicate each phase of what we propose, as well as to identify the alteration of the walking pattern to send a warning about a possible fall of the person. Also, this system is driven to be developed in three modules, as shown in Figure 1.

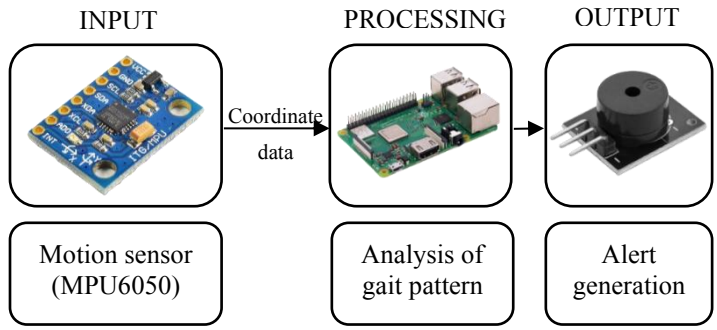

Fig. 1 System block diagram

In the first stage, a MPU6050 essential device is used to detect movement by observing the person's gait instantaneously without inconvenience, establishing communication through the I2C communication protocol, in addition to having a motion sensor that has a gyroscope that gives us the advantage of being portable and low cost. In the processing stage is the Raspberry Pi 3 board that is used to process the data of the motion sensor that is connected in one of its ports, in such a way that the variation of the walking variable of the sick person affected by this can be detected using the algorithm developed in Python. In the output stage is the alert generator that will be activated when an abnormal gait is detected in the patient by means of an auditory signal, for the person to take recommendations about what can happen if he/she does not follow the indications to avoid suffering an accident due to the fall.

IV. OPERATION OF THE SYSTEM

For operate the proposed equipment, the variation of walking of a patient with Parkinson's disease must be considered, for this the motion sensor was placed in position 1, as shown in figure 2. The values of the variation of the gait by the gyroscope (sensor) were read by means of the I2C communication protocol where the fixed point of the gyroscope gives a resistant performance in the course of time.

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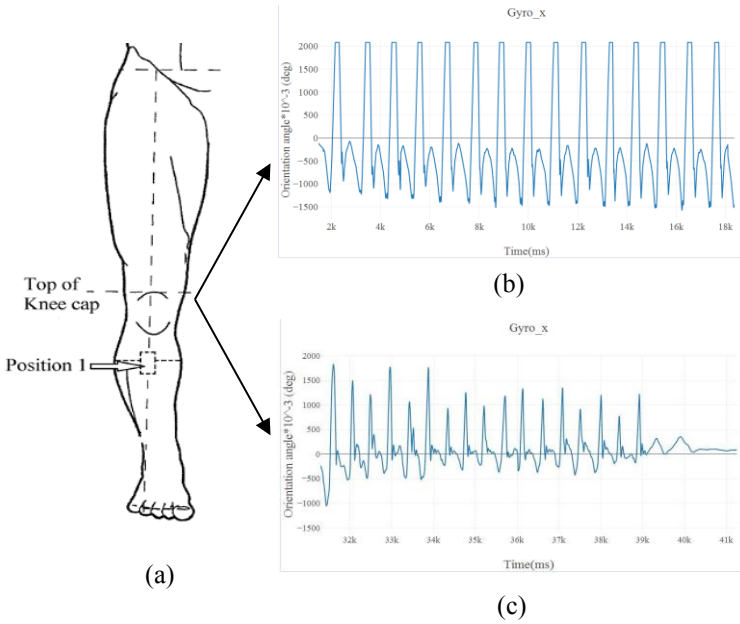

Fig. 2. (a) Position of the motion sensor. (b) Normal rate pattern. (c) Abnormal pattern of rate

When you turn on the system, the MPU6050 motion sensor starts and starts running. The three-dimensional gyroscope (sensor) coupled to the lower knee and the top of the tibia, measures the angular rate at a ±250deg/sec for each axis of rotation, although the main data captured for analysis are on the x-axis. The processor will then read the sensor data through the communication is made to acquire the sensor data and to achieve the targeting angle values from the angular rates.

According to the half positive cycles that are obtained from the gait pattern, the maximum orientation angles (P_k) are considered to calculate the values of the standard deviation (σ_n) with the following formula shown below.$P_k \sigma_n$[16]

$$\sigma_n = \sqrt{\frac{1}{x+n}\sum_{k=1}^{x+n}(p_k - \mu_{x+n})^2}$$

Where it is in the range of 1 to N, with N being the list size of the standard deviation, X the minimum number of samples needed to initially analyze the variable of walking and to be able to observe the average of all. $(\mu_{x+n}), (x+n)$

the standard deviation ratio is demonstrated as long as the following is met.[16]

$$\sigma_n > \sigma_{n-1} > \sigma_{n-2}$$

According to a growth, it will be determined that patients with Parkinson's are deviating from average values of the gait, therefore, the system will generate an alert by means of an auditory signal.

V. RESULTS

According to what has been developed, the system fulfills the objective of helping patients with this disease to prevent patients from being affected with Parkinson's disease is met, proactively helping them to detect and prevent falls by analyzing the gait pattern, allowing them to perform their activities safely.

With the development of the algorithm, variations in variable gait in sick patients were analyzed and detected. According to the prototype developed, as shown in Figure 3, in normal gear the BUZZER will remain off, while, if the algorithm detects abnormal gears, the BUZZER will light up through a blue LED and emit an auditory signal to prevent the patient from suffering a fall.

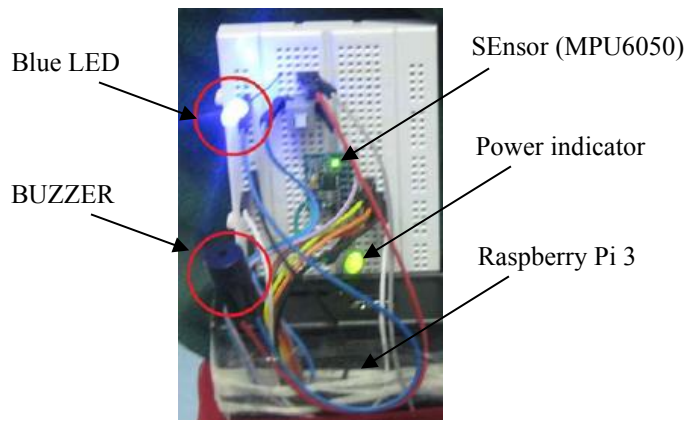

Fig. 3 Prototype of the automatic system

The development of this work is effective and adaptable to implemented in any medical center for the efficient treatment of Parkinson's patients. In addition, this non-invasive system does not generate discomfort and works at 97.99% efficiency unlike other systems.

This system is viable since it is not expensive due to the electronic components that are used for the implementation of the system, so its development is feasible since they are components that are easily found.

According to the gait pattern of patients with Parkinson's disease, the algorithm developed in the system can determine the variability of gait with the parameters of the mean and standard deviation, causing an alert to be generated if an abnormal gait is detected. So, this system is important in this current situation where it is necessary to detect the disease in time.

VI. DISCUSION

Now there is too much research on gait paralysis in sick patients, therefore, this research work differs from others, for example, the work carried out by [12], the researchers decided to develop a portable system to identify gait paralysis to be adapted to the sick patient by acoustic alerts. where their result was low efficient 90.2% and is not used in several places in some patients because it is a bulky and uncomfortable system to use.

The research conducted by [13], the researchers developed a system to identify the walking rhythm and alert when there is a paralysis. They also obtained an efficiency of 91.11%, but this

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project is limited presents a delay problem in alerting the patient with Parkinson's, alerting him after 3 seconds and generating that the patient suffers some fall.

The research conducted by [14], the researchers developed a system that is based on detecting the walking variable and helping to identify the paralysis of the same for sick people. They obtained an efficiency of 93.80%, but this project is limited when evaluating their system in a brief time, it was observed that the parameters of the march do not detect them instantly.

The research conducted by [15], the researchers developed a smartphone-based to freeze gait monitoring in patients. They obtained an efficiency of 91.79%, but does not have good viability when detecting the freezing of the gait, so the algorithm does not discriminate daily activities.

VII. Conclusion and Recommendation

It follows that the automatic fall detection presents results that demonstrate the correct standard deviation approach to detect gait variations in Parkinson's patients.

It is concluded that the system works as fast as 0.99 seconds to alert the sick person to be attentive and not to fall to the ground.

It follows that all the data analyzed from the prototype are fundamental for the person, so that the prototype is important for all people and they do not spend money on equipment that is not efficient.

Future, a vibration alert will be added to this prototype for quick sensing.

It is recommended that the patient place the system in the part indicated in the methodology so that it can detect the variation of the gait efficiently.

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